

A Packing of 55 Lines in $\mathbb{R}^{15}$

Limin Ge*

October 8, 2026

Abstract

We give 55 1-dimensional subspaces of $\mathbb{R}^{15}$ whose smallest squared chordal distance is at least 0.933025. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 0.932298, obtained by deleting 1 of the 56 lines of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 55 subspaces of dimension 1 in $\mathbb{R}^{15}$ whose pairwise squared chordal distances $D(a, b) = 1 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 0.933025 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 15$, $n = 1$. The cell $N = 55$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 0.932298, obtained by deleting 1 of the 56 lines of the packing found by this search. The Rankin simplex bound for this cell is $1(15 - 1)/15 \cdot 55/54 = 0.950617$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m15n1N55.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 69 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:
933025186918791223063028045665224719294290528311
denominator:
999999999999370685948261966090832066912472201912

B Coordinates

55 blocks of 15 rows with 1 integers each; divide by 10^{12} . File: `grassmann/packings/m15n1N55.txt`.

-31350384038	200707960966	-256387813356	222076569998
-203020196038	-19902690594	-453536628906	-327735755968
9613697653	355167007541	-123140488102	598447971355
-99984919192	-407913299	-360542677597	174746738825
-199675823849	195238793001	31478244230	-311691690020
-98359909773	-26544555181	-113279232102	-147180363685
-347075188058	-28377413486	416970140472	369015079219
724116787137	-232371057092	70784188437	248749353633
330455046707	226039196635	-29363280472	162461296714
302391990462	463653461860	-114546603699	-8063836200
-94744329780	-172576609202	-15731658688	-190810342553
70479151744	183093796949	317206469311	-147891029772
-116926920609	184288018287	-95745040517	-170008085489
-140041957662	-241787170320	-181588259147	55453126080
740960511165	177600693801	-118082666102	144968600884
199677961588	-267747491755	105086420544	199298464990
92328684915	130595082227	221142970326	674384755224
-29532985938	350076691102	189080985256	-288549239771
311378759288	-167862441075	-687093996173	-14648893807
-61230893977	-501676315677	115731974928	-123557347675
-60255745907	81854437577	381852310827	-292979342694
-119221180907	-430565851535	-218251581349	-10359555549
83268858737	82329043848	138193506112	174784366252
22104754183	398059426992	69760429936	56822217686
421842262485	35501696953	-210826245074	162006166261
-194719373397	48519988526	350813514440	-290308610116
22272183469	-151359153926	167113563988	303448838341
-610665087439	-259458032149	-229152031072	198884324431
381872388289	-196561965024	298331350319	159562476313
299429887346	-161578343278	-182751193286	93674625207
3865601024	-15847633986	-152420841232	270269575018
150031906463	-54228792337	180578829262	-279366664332
478285126217	135604548847	581861359559	-161825511922
8527023475	435112092993	-243011901808	76889612835
87139594809	522679339841	-182117234449	-41852593775
568491812168	154092628718	-180797242251	303347731566
165275921160	-118240745724	-187663736031	-764351393886
1554633446	-279514925059	-183159191390	-138443775933
-187445971852	-27067148092	-284422742599	38245331038
123116790975	127299862998	-71546083378	2671887253
-111296280964	143143353522	334423769158	100608323517
-25680801912	-284334614871	-245367980059	-84452666716
548393125882	-68308299196	-109741547413	-237398311090
135092570401	-176553483935	269964389822	-207933887868
90698347153	1592875872	-73102452804	-36322234119
124822887550	-666293679499	69053319752	182175691492
-168686607885	25143185830	333609936436	-33058363007
226369830465	360337202986	-63609734486	-455057077948
114794632369	-234302759027	310746153284	-615962870445
-87499608488	317351685735	-649897677369	-133407469629
-435240066006	28629385872	66518729401	235399193894
-42666923222	220916232085	43760853761	-46925300897
272703717272	-319810480441	-39748566466	212201232742
-383938467082	-165081179383	4781947349	-224010630972
66543254244	550595102063	-301171825258	57946691995
28687894409	-226825767220	207752334324	-97395201625
-174110427729	151345417391	-165056152512	-13522296828
606688711141	452904335621	-170016913888	-379124052518
-152253675216	17928710785	65873538493	-117846457923
209853581438	223817390732	-472985605779	194833736996
613147731385	323221456030	61628044070	158981842939
-92099658578	5027538344	382218919143	71281333931
-211017255510	-192199975359	-300722891821	-510591104383
22467631926	182792562986	-221758893742	630296181187
-466695493817	113781234584	145252453323	-180490765681
349256270150	-395970861610	-9568674718	-69222693763
19875041517	-455629340597	-268853556459	-86221325803
27776321497	-79045269921	-460652150242	-459216260059
-142565987460	77387248679	4841262783	142134927261
-65699687060	-10322920618	-278277708877	97415051659

-71459163804	-479604241410	-332444360073	-63767196085
43935162929	204991387307	-463612689882	-164410505064
103304141891	318205849700	-102404997448	-213772579785
-89228734078	-160645578372	283761647732	13385544006
24950984202	-73572983389	300526811131	627149053318
-420619745740	-217033811044	-476750904112	-152640654334
280472291069	-227279497832	-41164990611	297093053764
-86884239500	294470285306	-84824456158	202627267496
95905053230	-66374434234	-3906687467	166148732230
-426802275416	36631453186	-94701409629	45832882033
48893849879	101605119966	-2470103960	-240652407729
-8401951084	455118998264	228199363710	89490139174
-132919550537	250386126985	-135623881135	216504864387
-122124129732	309943779931	-129133974032	981046562
-120029031024	385498846078	389347248412	19312011167
185143824504	-188681399617	154593298512	267413409013
216479075211	69758834241	50324568864	-1570523077
460284357227	129081624702	334825460067	481572611374
-38557655195	-548118220290	317075637335	-194372757023
448962808242	-222626496394	246250348943	444625936596
-252633480656	-200122600790	326974058579	-311518975444
-203058585533	-149149346736	-106412909853	289055224027
-425855625671	-2862492272	274247592445	268518213751
119021142263	-88042332494	310012072056	-217621407297
-270141417174	577469149406	25672139884	-230627948947
236804645120	54363049713	413925584164	-127882997133
-34951705005	191483833646	429293768042	71360625114
161607012367	-38028307212	-73123165632	101136212875
-203519225824	-13404969135	333593770528	275331054150
285680644257	-187707724976	-188572864812	73143206935
83631775472	-161894529590	25742808339	-234939979058
-90928535618	-145565670672	151525499166	-3413995754
167642976784	26969348177	178623129219	537602425624
-609572120915	-198035798592	-82524723979	24655736970
-71099220925	-462118137302	-321976173663	-122234718634
-607155935327	335540923497	135500379754	313438879432
-150191560964	-376340499814	-135867233264	-30472445674
-187919116223	-164859672660	-125279153947	-446188574463
-70231898251	-374800158079	-23764491865	14190981777
269398782074	-465018914193	-666646666416	436961750736
414808021759	-81647979177	-3130994289	3158777973840
205572050109	209208076162	27170371188	-68090469772
221759706561	-10445884142	265473786100	-202628392698
-202783312929	-241058714514	697167561249	-2974847316
-20389524794	240391184499	35877014815	-105628572444
-211699801045	247938514566	-55441412025	290038873985
247371136011	-127121452982	255512200775	-166772616174
-18480768366	512727682985	-333208685001	347685337665
36970386400	67239807762	850215911	185068914513
288487399745	-233074341513	207798831443	-114168570748
15962836867	-27835837303	250664051834	304514957259
290542936579	272485651896	-72708024954	-76328510662
171461763509	-49500958857	9465235266	165694493606
285358540831	4734016799	-305149181430	-392649128490
198457562292	233964335184	-93751699300	72553348412
-316621394663	-69738652680	-220844770785	141711808533
-529852096189	205955500485	-121378837661	-332423445681
299824465519	297679371948	-34274471900	-13612663498
-351213782147	45289286119	176447279874	540848047167
-18202560326	-260873726878	-289350622393	75066211823
78774314586	185681316689	-279577484828	15552544280
-86742974432	-144575416961	465211992506	328818192784
-72384742684	150159384560	13323878688	68206243482
297075980052	-150747506991	-87833019590	-475395325079
-251640640602	504979000723	-251483653935	-244878387965
-164405930324	-341986192385	32582973464	62175893829
-163923881786	-507518571115	284997938995	-211247174435
50846667011	95143053460	582495118918	-324530389427
130509533338	18756097729	-96021894591	303999496485
426070351127	-386613816689	-218059207802	73916484718
350369257014	41331108655	-158609553697	315495337124
-317771107943	-21200517154	266237254144	109097108530
-129625839746	-204798155143	120212474048	478033301626
-7521908348	320481825	85588626383	-61136712672
-410509837075	-93312235955	-276879603144	415262049204
-325314520065	232801239449	225491798396	22437294763
-51838491014	-85143280783	-62531619922	61043665342
106912873460	-517306376491	166928265708	-358491276721
295525412829	28075072848	-60402905680	485707883372
-358536371624	247116409541	-463010626872	61152461779
300749799313	-30302477889	-27224856966	-261053157399
-118880610567	-481206050889	36192236932	-104525591252
-128910303949	417117210107	460433915861	-188081226935
-119963863829	-159133926000	117434858866	-113266913310
82749393002	441045712499	56575764857	184213125718
78514864283	260049468707	546797568462	-439102783742
81340792215	-85927701891	-101645814686	294248781110
442263425697	-344314208142	-123530838793	99329325014
-299674255013	172055373156	-204993837985	68463274973
43410327238	114234299793	94636778527	-158956383371
-354524436413	59485834540	368011406864	245694555628
-64323552433	-433204906424	-165062526388	-430327103550
152802633914	-150072768304	-72664802837	25864233096
635752978760	-450291465030	-111930648050	-314885540336
-11736670660	11783587056	-634708066046	409678711225
263426423507	-286921190269	-34568830705	301858013043
272381719396	57563657775	-123227636547	-30185731399
-213009351379	204145743463	-368258616980	-17542091958
-238129639636	-247584871620	-354410060042	-144559706119
362459014917	347948707389	-191977823707	-326576462109
-179986429210	256165421140	16010662132	-9716506415
23985769289	7875036031	-163155431639	295060859693

158662986460
-359598396319
-261580746684
-452153279732
-98553973613
-290131549171
-133379015702
-163515637339
186242779074
125034401487
-76111372328
-284359000878
-583883771809
-203087609230
-109523139750
-98636654271
-218755793447
-326330078187
-55831253564
-129955544099
-125357537497
-213650313272
-475763112579
23494709748
-211649630188
339844480437
342156466296
-249956266919
-2424604358
446396533384
-193389958872
-93408559299
264703446162
19062916396
-412129472838

198624887074
-179439717682
126176141201
-119165343198
280248790123
-18886954873
263782691382
245904439476
349933666676
-153035503163
202544649349
-511959528102
31494872480
-45204040015
187200426307
-343534068745
98810458930
57332400055
192653844602
-12860646701
79248213337
416939793427
-594523875084
490505516039
-63551472691
59531653623
-108707972932
-457645542205
91937477864
17027308418
457228895835
-135132885324
414696317541
-213150136489
-255323578031

-342672940970
188148052292
165015084000
-90657938544
-222183747413
165516780262
-82874162426
-544516955262
268030963904
-388679195923
-212535469707
-114374251146
-30645874131
256882306523
-135519982112
75063123919
97973141215
103784593912
-499457942468
-186841698668
-22484582193
161755855269
345514533189
-205788918908
-354057473145
-333094614652
-159104575548
-87690051693
-187247475662
-470804838508
12949736290
-153318659259
-246527495655
228320224493
24538855839

344308268546
-237715607756
-120384609121
185638123055
-176134834795
-188489275135
132452985493
-162610309475
80038822885
224938406666
-256003118541
-533867355190
-85338433741
521801996171
-190837510660
159144187146
-302732871212
-76904020751
-24294504381
-322186977081
26739131540
-622016790068
-203810781208
-419374921207
73625646231
-226887335770
60049468176
-222400562702
-347657407846
32683344181
210074309262
92873223426